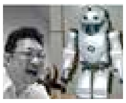

Sony can manufature robots?
Sony has claimed that it can manufacture humanoid robots once their purpose is clear. **https://dgit.in/dec22-05**

Pricier apps on the App store?
Apple has announced a major overhaul to the app store pricing which means apps could get pricier. **https://dgit.in/dec22-06**

Emile Berliner

The microphone worked on Reis' concept of audio propagation even further by using metal plates instead of using parchment or liquid. The metal plates had carbon granules between them and as the thinner plate would move, the resistance between the two plates would vary because of the movement of the connected carbon molecules producing the desired audio signals that were needed to transmit audio to other equipment. Now, even though Berliner is credited with making the Carbon Microphone, Sir David Edward Hughes also made a similar invention in the same year and actually demonstrated it before the American duo of Edison and Berliner. However, he did not get it patented calling his invention "a gift to the world". He was also the first man to bring the term "microphone" to his invention, perhaps as a tribute to Sir Charles Wheatstone.

Note: The patent to the original Carbon Microphone, held by Berliner, was bought out by Alexander Graham Bell for $50,000 (roughly $1.1 million in today's times), but after a ton of legal trouble, in 1892, the United States Supreme Court ruled that it was Edison who had actually invented the microphone and the original patent was awarded to him. While we are not sure if Edison used his influence over the world at the time or not, given his past adventures, we simply cannot disregard the possibility.

THE CONDENSER MIC

It was in the year 1904 that the vacuum tube was invented. However, it wasn't till 1906 till Lee De Forest developed the triode that was used immensely in the microphones that came up next due to its amplifying nature. This invention was one of the biggest reasons that the condenser microphone was able to come into existence as the initial models had very weak outputs and needed the amplifying powers of the triode.

It wasn't until 1916 that we got to see the first-ever condenser microphone come into existence at Bell Laboratories(Then Western Electric) by the inventor, E.C. Wente. Much like the carbon mic, the condenser also came equipped with two plates out of which one moved. However, the difference was that there were no granules in the middle of the two plates. Instead, a steady current was constantly applied between the two and while the moving plate would vibrate, the capacitance between the plates would change resulting in the production of the audio signals. Unfortunately for us, there wasn't a lot of juicy controversy surrounding this invention like the others but it's like they say, the 1800s was a wild time for science and its followers.

The first ever Condenser Microphone

A WEIRD TIME TO TEST CRYSTALS

French Physicist Paul Langevin was the first ever person to use crystals(specifically piezoelectric ones) to detect sound. What exactly was he up to with so many crystals? We do not know and honestly, we hope we don't get to because we'd rather not mix into bad business deals or end up like Walter White. Either way, the crystals were initially used in ultrasonic submarine detectors and not for microphone-related purposes. They were a part of early submarine systems instead and were used as precursors to the more advanced SONAR systems that we have today.Then, in 1919, Alexander Nicolson made the first ever piezoelectric microphone for capturing sound waves while also developing the piezoelectric loudspeaker and the phonograph pickup.

Paul Langevin

Piezoelectric Crystals

THE CONCLUSION(?)

Now we may not have reached current times just yet. But for the most part the foundations of the microphones were laid out during the times we talked about in this article. If you decide to take a deep dive into the world of microphones, you'll be surprised how things have changed, but the fundamentals have stayed the same.